

## *Adherence to Mediterranean diet and intake of antioxidants influence spontaneous conversion of atrial fibrillation.*

. . . . .

BACKGROUND AND AIM: The Mediterranean diet (MedD) has long been associated with lower incidence of cardiovascular disease. Little information is available on association between MedD, vitamins intake and arrhythmias. We sought to investigate the relationship between adherence to MedD, antioxidants intake and spontaneous conversion of atrial fibrillation (AF). METHODS AND RESULTS: A group of 800 subjects was included in a case-control study; 400 of them had a first detected episode of AF. Nutritional parameters were assessed by a self-administered food frequency validated questionnaire and completed by an interviewer-administered 7 days diet recall. Adherence to MedD was evaluated using the Mediterranean Score and intake of antioxidants from food was calculated. Adherence to the Med Diet was lower in patients that developed AF compared to control (mean Med Score:  $22.3 \pm 3.1$  vs  $27.9 \pm 5.6$ ;  $p < 0.001$ ). The median value was 23.5 (Q1-Q3 range 23-30) in patients with AF and 27.4 (Q1-Q3 range 26-33). The estimated intake of total antioxidants was lower in patients with AF ( $13.5 \pm 8.3$  vs  $18.2 \pm 9.4$  mmol/d;  $p < 0.001$ ). Patients in the highest quartile of Mediterranean Score had higher probability of spontaneous conversion of atrial fibrillation (OR1.9; 95%CI 1.58-2.81). High levels of antioxidants intake were also associated with an increasing probability of spontaneous conversion of arrhythmia (O.R. 1.8; 95%CI 1.56-2.99;  $P < 0.01$ ). CONCLUSIONS: Patients with atrial fibrillation had lower adherence to MedD and lower antioxidant intake compared to control population. Moreover patients with arrhythmia showing a higher Med Score had more probability of a spontaneous conversion of atrial fibrillation. Copyright © 2011 Elsevier B.V. All rights reserved.